IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EEE/EIE PART II: MEng, BEng and ACGI

SIGNALS AND LINEAR SYSTEMS

Monday, 8 June 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Question One carries 40% of the marks. The other 2 questions each carry 30%.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s): P.T. Stathaki
Second Marker(s): J.A. Barria

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1. This question carries 40% of the mark.

(a) Consider each of the following signals in continuous time. For each case, specify if the signal is causal/non-causal, periodic/non-periodic, odd/even. If the signal is periodic specify its period.

(i) $x(t) = |\cos(2\pi t + \pi/6)|$ **[2]**

(ii) $x(t) = e^{-t}$ **[2]**

(b) Consider the signal

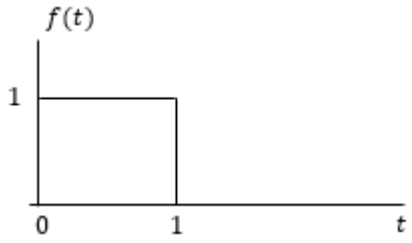
$$x(t) = \begin{cases} 1 - 3t, & 0 \le t \le 2 \\ 0, & \text{otherwise} \end{cases}$$

Now find each of the following signals.

(i) $x\left(-\frac{t}{4} + 1\right)$ **[2]**

(ii) $x(3t + 4)$ **[2]**

(c) Find the system's zero-state response given that the impulse response is $h(t) = e^{-3t}u(t)$ and that the input $f(t)$ is as shown in Figure 1(c) below.



[6]

Figure 1(c)

(d) (i) Use the time-shifting property to show that if $x(t) \Leftrightarrow X(\omega)$, then $x(t + T) + x(t - T) \Leftrightarrow 2X(\omega)\cos T\omega$, where T is a constant parameter. We denote with $X(\omega)$ the continuous-time Fourier transform of the signal $x(t)$. **[3]**

(ii) Use the result in (d)(i) above to find the Fourier transform of the signal shown in Figure (d)(ii) below.

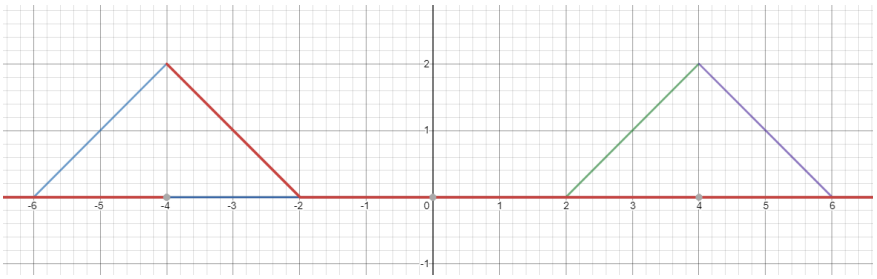


Figure (d)(ii)

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Use the fact that $\Delta\left(\frac{t}{\tau}\right) \Leftrightarrow \frac{\tau}{2}\text{sinc}^2\left(\frac{\omega\tau}{4}\right)$ with
$\Delta(x) = (1-2|x|)(u\left(x+\frac{1}{2}\right) - u\left(x-\frac{1}{2}\right))$

[3]

(e) The output $y(t)$ of a continuous-time LTI system is related to the input $x(t)$ by the differential equation

$$\frac{dy(t)}{dt} + 3y(t) = x(t)$$

Sketch the Bode plots of the system. Detailed explanation is required. **[5]**

(f) Figures (i) and (ii) below show the Fourier spectra of signals $f_1(t)$ and $f_2(t)$. Determine the Nyquist sampling rates for the following signals.

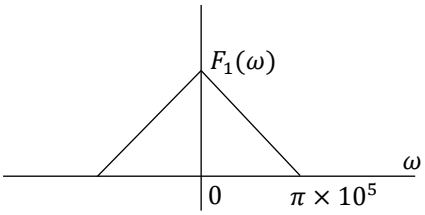


Figure (f)(i)

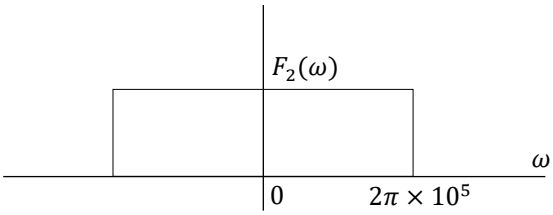


Figure (f)(ii)

(i) $f_1(t)$ **[2]**
(ii) $f_2(t)$ **[2]**
(iii) $f_2^3(t)$ **[2]**
(iv) $f_1^2(t)$ **[2]**
(v) $f_1^2(t)f_2(t)$ **[2]**

(g) Consider the discrete-time, causal LTI system with input $x[n]$ and output $y[n]$ related with the difference equation:

$$y[n] - \frac{5}{8}y[n-1] + \frac{1}{16}y[n-2] = x[n] - \frac{1}{8}x[n-1]$$

(i) Find the analytical expression and the Region of Convergence (ROC) of the $z-$transform of the impulse response of the above system. **[3]**

(ii) Find the analytical expression and the ROC of the $z-$transform of the output if $x[n] = \left(\frac{1}{4}\right)^n u[n]$. **[2]**

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2. This question carries 30% of the mark.

(a) (i) Find the analytical expression and the Region of Convergence (ROC) of the Laplace transform of the continuous-time signal $x(t) = ae^{at}u(-t) + e^{-at}\delta(t)$, with a real and positive and $u(t)$ the continuous-time unit step function. **[5]**

(ii) Explain whether the Laplace function is adequate in order to determine the corresponding function in time. **[2]**

(iii) Find the analytical expression and the ROC of the Laplace transform of the signal $\delta(t)$, with $\delta(t)$ the continuous-time impulse function. **[3]**

(iv) Find the analytical expression and the ROC the Laplace transform of the signal $u(t - a)$, with with a real and $u(t)$ the continuous-time unit step function. **[2]**

(b) Consider a continuous-time causal signal $x(t)$ with Laplace transform $X(s) = \frac{5s+11}{s^2+5s+4}$. Find the analytical expression of $x(t)$ and furthermore, find the ROC of $X(s)$. **[5]**

(c) Consider a continuous, Linear Time-Invariant (LTI) system with input the unit step function $u(t)$ and output the signal $y(t) = -\frac{2}{3}e^{3t}u(-t) + \frac{1}{3}e^{-3t}u(t)$. Determine the transfer function of the system $H(s)$ and its ROC. **[5]**

(d) The output $y(t)$ of an LTI system is related to the input $x(t)$ by the differential equation

$$\frac{d^2y(t)}{dt^2} + 9\frac{dy(t)}{dt} + 14y(t) = 3\frac{dx(t)}{dt} + 16x(t)$$

Let $X(s)$ and $Y(s)$ denote the Laplace transforms of $x(t)$ and $y(t)$, respectively, and let $H(s)$ denote the Laplace transform of the system's impulse response $h(t)$.

(i) Determine the transfer function $H(s)$ of the system. **[3]**

(ii) Determine all possible forms of the impulse response $h(t)$. **[3]**

(iii) Can the above system be both stable and causal? **[2]**

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3. This question carries 30% of the mark.

(a) Find the Inverse Fourier transform of the function $X(\omega) = \frac{\pi}{W}\text{rect}(\frac{\omega}{2W})$, where W is a constant parameter and **[3]**

$$\text{rect}(t) = \begin{cases} 1 & |t| < 0.5 \\ 0.5 & |t| = 0.5 \\ 0 & \text{otherwise} \end{cases}$$

(b) Determine the Nyquist sampling rate and the Nyquist sampling interval for the following signals.

(i) $x_1(t) = \text{sinc}^2(50\pi t)$. **[3]**

(ii) $x_2(t) = 0.02\text{sinc}^2(50\pi t)$. **[3]**

(iii) $x_3(t) = \text{sinc}(60\pi t) + 3\text{sinc}^2(100\pi t)$. **[3]**

(iv) $x_4(t) = \text{sinc}(60\pi t)\text{sinc}(100\pi t)$. **[3]**

(c) (i) Consider a discrete signal $x[n]$ with $z-$ transform $X(z)$. Find the $z-$ transforms of the signals $x_1[n] = x[n-1]$ and $x_2[n] = x[-n-4]$ as functions of the $z-$ transform $X(z)$. **[3]**

(ii) Consider a signal $y[n]$ which is related to two signals $x_1[n]$ and $x_2[n]$ by

$$y[n] = x_1[n] * x_2[n]$$

where

$$x_1[n] = \left(\frac{1}{4}\right)^n u[n] \text{ and } x_2[n] = \left(\frac{1}{5}\right)^n u[n]$$

with $u[n]$ the discrete unit step function. Determine the $z-$ transform $Y(z)$ of $y[n]$. **[3]**

(d) Consider an odd sequence $x[n]$ with $z-$ transform $X(z)$.

(i) Find $X\left(\frac{1}{z}\right)$ as a function of $X(z)$. **[3]**

(ii) From your result in part (d)(i), show that if a zero (pole) of $X(z)$ occurs at $z = z_0$, then a zero (pole) must also occur at $z = 1/z_0$. **[3]**

(iii) Verify the results in part (d)(ii) for the sequence $u[n] - u[-n]$, with $u[n]$ the discrete unit step function. **[3]**

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